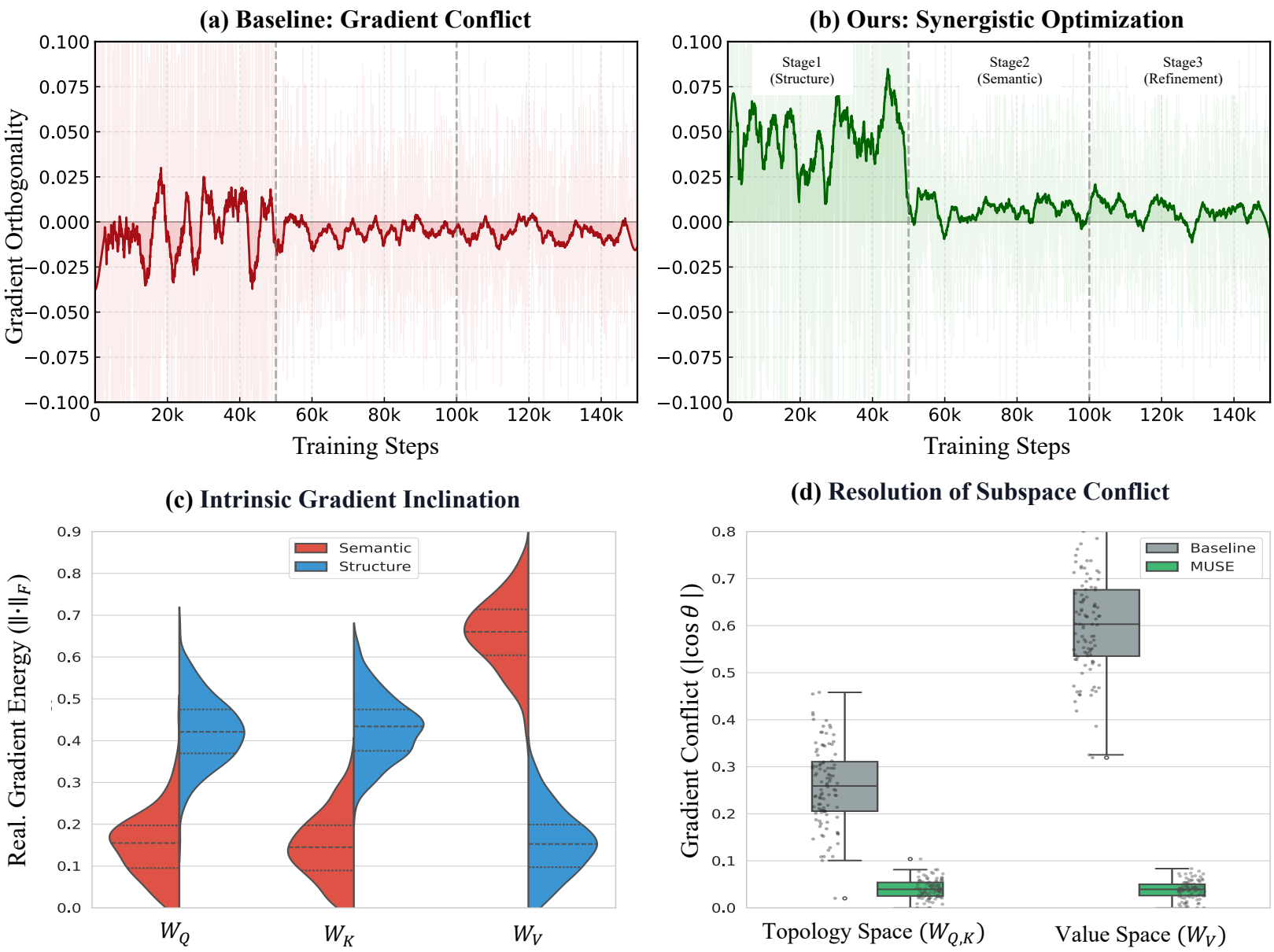
(a) Baseline: Gradient Conflict
Gradient Orthogonality
Training Steps
(b) Ours: Synergistic Optimization
Stage1 (Structure)
Stage2 (Semantic)
Stage3 (Refinement)
Training Steps
(c) Intrinsic Gradient Inclination
Real. Gradient Energy ($\|\cdot\|_F$)
Semantic
Structure
W_Q
W_K
W_V
(d) Resolution of Subspace Conflict
Gradient Conflict ($|\cos\theta|$)
Baseline
MUSE
Topology Space ($W_{Q,K}$)
Value Space (W_V)